

Versioning Agents and Managing Dependencies



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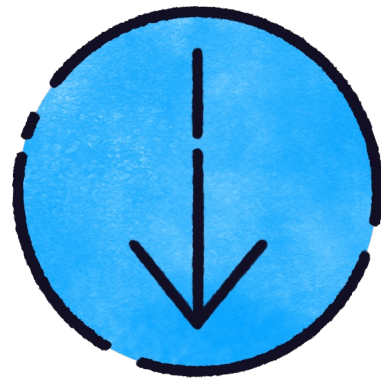
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Why Versioning Matters for AI Agents

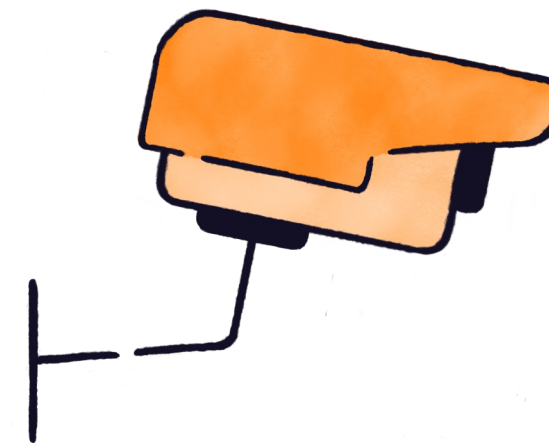


The Cost of No Version Control



Agents silently degrade

It happens when prompts are deprecated or models are upgraded
(GPT-3.5 → GPT-4)



No visibility

If no visibility of the change is available, it will lead to production chaos



Why Agent Versioning Is Different

Traditional code

- Git handles versioning
- Tests catch issues
- Changes are deliberate

vs.

Agent systems

- Prompts subtly change behavior
- Model upgrades can alter outputs without code change
- AI agents need explicit version tracking beyond just code



What Should Be Versioned?

Prompt versions

**Model used (GPT-4,
Claude 3, etc.)**

**Tool configuration
(APIs, plugins)**

**Environment (vector
DB, memory settings)**

Release notes



Benefits of Versioning

Trace issues to exact
prompt/model/tool
config

Easier handoff and
collaboration

Clear release history
for audit trail

Rollbacks possible

Compare outputs
across versions

Safer
experimentation



Demo: Creating an Agent Manifest



Tracking Changes with Git and Tags



Git is not just for code! It works great for: prompts, configs, manifests, and test outputs.



Example Git Repo Structure

```
agents/  
  globomantics_assistant/  
    prompts/  
    manifest.yaml  
    tests/
```




```
// Using tags to mark releases
```

```
git tag -a v1.4.0 -m "Improved prompt  
and updated to GPT-4"
```

```
git push origin v1.4.0
```

Bookmark versions

Roll back failed updates

Share releases with QA teams



Dependency Mapping: APIs, Models, and External Services



Examples of Common Agent Dependencies

**APIs (Salesforce,
Slack, Notion)**

**Vector DBs
(Pinecone, Weaviate)**

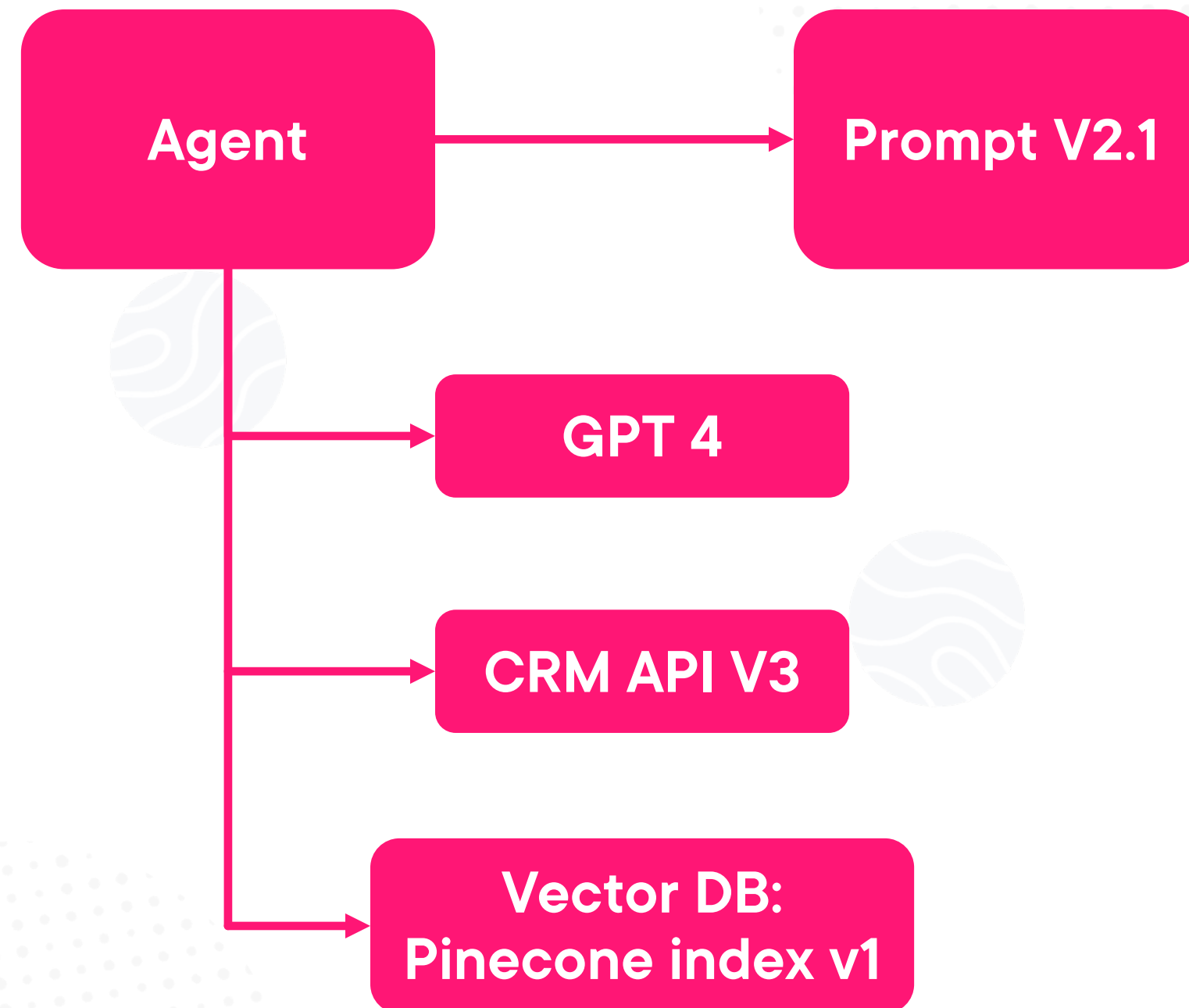
**LLM models (GPT-4,
Claude, Mistral)**

**Authentication
providers**

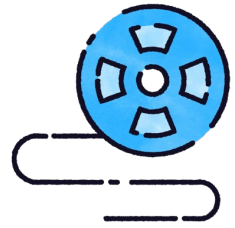
Retrieval indexes



Visually Mapping Dependencies



Change Management Strategy



Record version



Regularly check for breaking changes



Subscribe to API deprecation updates



Build alerts on 5xx errors or latency spikes

